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MAT 213 Brief Calculus

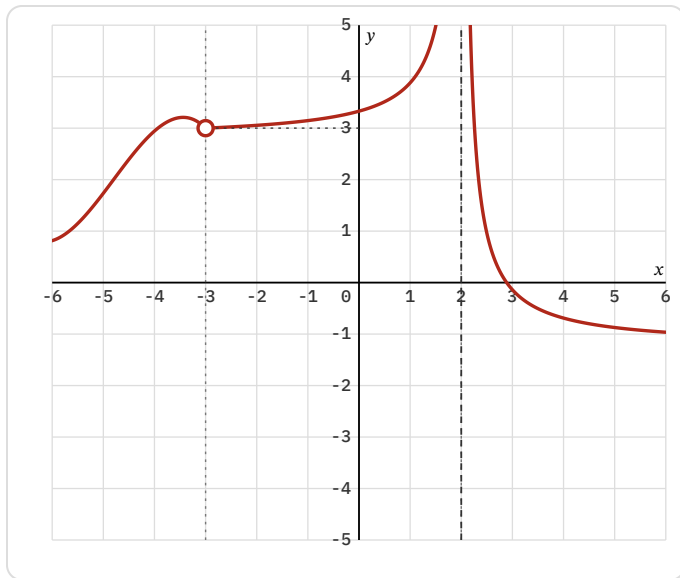
Mock Midterm D — Answer Key

Practice midterm (AI-Free) · Sections 1.1–1.4

Total: 60 points

1.

Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write DNE.



a.) $\lim_{x \rightarrow -3} f(x)$ (2)

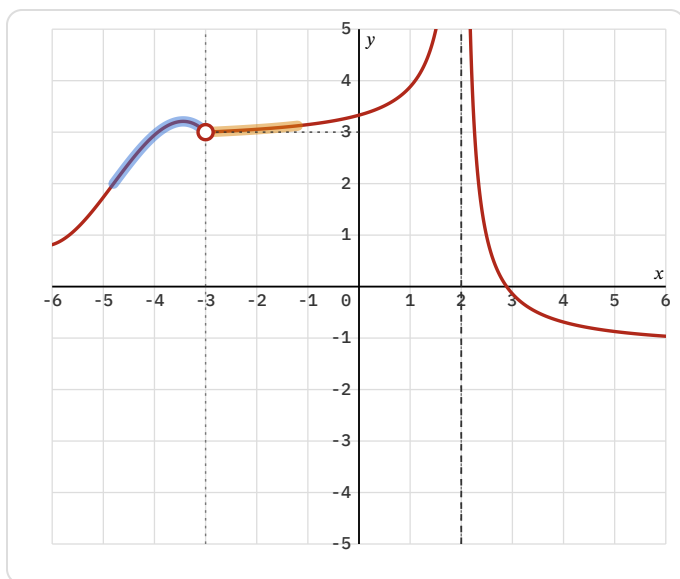
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b.) $f(2)$ (2)

DNE

4 pts

WORKED SOLUTION



— approaching $x = -3$ from the left — from the right. Follow each colored piece of the curve to the height it heads toward.

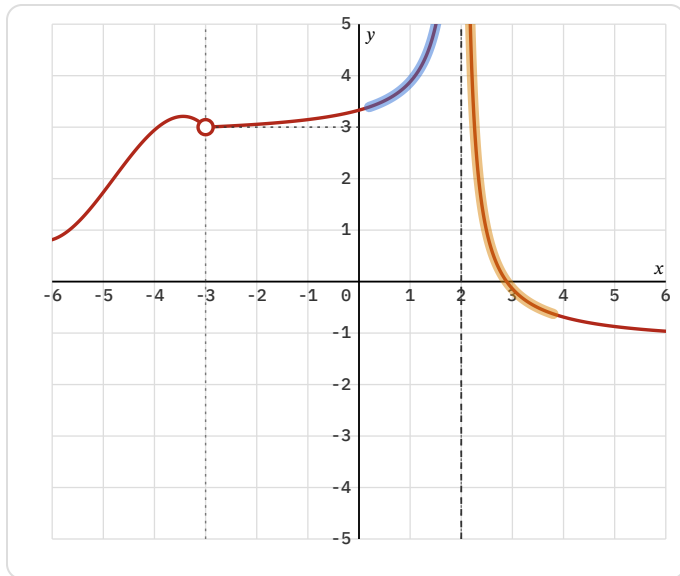
At $x = -3$

There is no filled dot on the line $x = -3$, so $f(-3)$ is undefined $\rightarrow$ DNE.

Tracing the curve toward $x = -3$ from the left, the heights approach 3 (open or closed dot doesn't matter).

Tracing the curve toward $x = -3$ from the right, the heights approach 3 (open or closed dot doesn't matter).

Both one-sided limits equal 3, so $\lim_{x \rightarrow -3} f(x) = 3$ — even though $f(-3)$ is undefined. The limit ignores the value AT the point.



— approaching $x = 2$ from the left — from the right. Follow each colored piece of the curve to the height it heads toward.

At $x = 2$

There is no filled dot on the line $x = 2$, so $f(2)$ is undefined $\rightarrow$ DNE.

From the left, the curve runs along the asymptote $x = 2$ upward, so the limit is ∞ .

From the right, the curve runs along the asymptote $x = 2$ upward, so the limit is ∞ .

Both sides go to ∞ , so $\lim_{x \rightarrow 2} f(x) = \infty$ (unbounded, so no finite limit exists).

2. Let $f(x) = \frac{-3}{x-1}$. Answer with a number, ∞ , $-\infty$, or DNE. 4 pts
- a.) $\lim_{x \rightarrow 1^-} f(x)$ (2)
 ∞
- b.) The vertical asymptote is $x =$ (2)
 1
-

WORKED SOLUTION

Near $x = 1$ the denominator $x - 1$ is tiny, so $f(x)$ is huge in size. Only the sign needs work.

Left of 1 (try $x = 0.99$): $x - 1$ is a small **negative** number, so $\frac{-3}{\text{small negative}} \rightarrow \infty$.

Right of 1 (try $x = 1.01$): $x - 1$ is a small **positive** number $\rightarrow -\infty$.

The sides disagree, so the two-sided limit DNE.

As $x \rightarrow \infty$, the denominator grows without bound while the top stays -3, so $f(x) \rightarrow 0$. The vertical asymptote is $x = 1$.

3. Calculate each limit algebraically. 4 pts
- a.) $\lim_{x \rightarrow 4} (-x^2 - 2x - 5)$ (2)
 -29
- b.) $\lim_{x \rightarrow 1} \sqrt{x^2 + 8}$ (2)
 3
-

WORKED SOLUTION

Each input is in the domain of its function, so direct substitution works.

a. $\lim_{x \rightarrow 4} (-x^2 - 2x - 5) = -1(4)^2 - 2(4) - 5 = -29$

b. $\lim_{x \rightarrow 0} \frac{x^2+9}{x-2} = \frac{(0)^2+0(0)+9}{0-2} = \frac{9}{-2} = -\frac{9}{2}$

c. $\lim_{x \rightarrow 1} \sqrt{x^2 + 8} = \sqrt{(1)^2 + 8} = \sqrt{9} = 3$

4.

Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE.

a.) $\lim_{x \rightarrow 0} \frac{-5x^2 - 9x}{3x}$ (2)
 -3

b.) $\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4}$ (2)
 $\frac{1}{4}$

4 pts

WORKED SOLUTION

a.

$$\lim_{x \rightarrow 0} \frac{-5x^2 - 9x}{3x} = \lim_{x \rightarrow 0} \frac{x(-5x - 9)}{3x} = \lim_{x \rightarrow 0} \frac{-5x - 9}{3} = -3$$

b.

$$\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4} = \lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{(\sqrt{x} - 2)(\sqrt{x} + 2)} = \lim_{x \rightarrow 4} \frac{1}{\sqrt{x} + 2} = \frac{1}{2 + 2} = \frac{1}{4}$$

Notation tip: keep writing **lim** on every line until the step where you actually substitute.

5. Compute each limit algebraically. Use $\infty / -\infty$ for unbounded results.

4 pts

a.) $\lim_{x \rightarrow \infty} \frac{-2}{5x^4 + 5x^3}$ (2)

0

b.) $\lim_{x \rightarrow \infty} \frac{-5x^4 + 6x^3}{6x^4 - 6x^2 - 1}$ (2)

$-\frac{5}{6}$

WORKED SOLUTION

a. Divide every term by x^4 , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{-2}{5x^4 + 5x^3} = \lim_{x \rightarrow \infty} \frac{-\frac{2}{x^4}}{5 + \frac{5}{x}}$$

The top $\rightarrow 0$ while the bottom $\rightarrow 5$, so the limit is 0.

b. Divide every term by x , the highest power of x in the denominator:

$$\lim_{x \rightarrow -\infty} \frac{2x^2 - 8x + 7}{5x} = \lim_{x \rightarrow -\infty} \frac{2x - 8 + \frac{7}{x}}{5}$$

The bottom $\rightarrow 5$, but the top is dominated by $2x^1$, which $\rightarrow -\infty$ as $x \rightarrow -\infty$; dividing by 5 gives $-\infty$.

c. Divide every term by x^4 , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{-5x^4 + 6x^3}{6x^4 - 6x^2 - 1} = \lim_{x \rightarrow \infty} \frac{-5 + \frac{6}{x}}{6 - \frac{6}{x^2} - \frac{1}{x^4}}$$

Every fraction term $\rightarrow 0$, leaving $\frac{-5}{6} = -\frac{5}{6}$.

6. Find each limit (end behavior). Use ∞ or $-\infty$ when the outputs grow without bound. 4 pts

a.) $\lim_{x \rightarrow -\infty} (x^2 - 5)$ (2)
 ∞

b.) $\lim_{x \rightarrow \infty} (-3x^4 - 5x + 4)$ (2)
 $-\infty$

WORKED SOLUTION

a. Leading term x^2 : degree 2 (even), leading coefficient positive. As $x \rightarrow -\infty$, x^2 is positive, times 1 $\rightarrow \infty$.

b. As $x \rightarrow \infty$, e^{-x} shrinks to 0, so the expression approaches -1.

c. $\frac{8}{x^2} \rightarrow 0$ as $x \rightarrow -\infty$, leaving -6.

d. Leading term $-3x^4$: degree 4 (even), leading coefficient negative. As $x \rightarrow \infty$, x^4 is positive, times -3 $\rightarrow -\infty$.

7. Let

4 pts

$$f(x) = \begin{cases} \frac{x^2 - 4x + 3}{x - 3} & x < 3 \\ 4x - 10 & x > 3 \end{cases}$$

Find each limit or function value algebraically. Write DNE if it doesn't exist.

a.) $\lim_{x \rightarrow 6} f(x)$ (2)

14

b.) $\lim_{x \rightarrow 3} f(x)$ (2)

2

WORKED SOLUTION

For $x < 3$, factor the top: $\frac{(x-3)(x-1)}{x-3} = x - 1$ (valid since $x \neq 3$ there).

a. $6 > 3 \rightarrow$ bottom piece: $4(6) - 10 = 14$.

b. $0 < 3 \rightarrow$ top piece, simplified: $0 - 1 = -1$.

c. Neither piece includes $x = 3$, so $f(3)$ is undefined $\rightarrow$ DNE.

d. Left: $\lim_{x \rightarrow 3^-} (x - 1) = 2$.

e. Right: $\lim_{x \rightarrow 3^+} (4x - 10) = 2$.

f. Both sides equal 2, so the limit is 2.

8. Is $f(x)$ continuous at $x = -2$?

3 pts

$$f(x) = \begin{cases} \frac{x^2 + x - 2}{x + 2} & x \neq -2 \\ -3 & x = -2 \end{cases}$$

a.) $f(-2)$ (2)

-3

b.) Is f continuous at $x = -2$? (1)

(A) Yes — $f(-2)$ exists, the limit exists, and they are equal

WORKED SOLUTION

1. $f(-2) = -3$ (given directly by the second piece).

2. $\lim_{x \rightarrow -2} \frac{(x+2)(x-1)}{(x+2)} = \lim_{x \rightarrow -2} (x-1) = -3$.

3. $-3 = -3$. All three conditions hold $\rightarrow$ **continuous**.

9. Classify each function at the given x -value.

2 pts

a.) $f(x) = \frac{x^2 - 25}{x + 5}$ at $x = -5$ (1)

(B) Removable discontinuity

b.) $f(x) = \frac{x+1}{x^2 - 2x - 3}$ at $x = -1$ (1)

(B) Removable discontinuity

WORKED SOLUTION

a. $0/0$ at -5 , but it cancels to $x - 5$, so the limit exists (-10) while $f(-5)$ is undefined $\rightarrow$ removable.

b. Factor: $\frac{(x-2)}{(x-2)(x-1)}$. The $(x-1)$ factor does NOT cancel, so $x = 1$ is a vertical asymptote $\rightarrow$ infinite.

c. Factor: $\frac{(x+1)}{(x+1)(x-3)}$. The $(x+1)$ cancels, leaving $\frac{1}{x-3} \rightarrow -\frac{1}{4} \rightarrow$ removable.

10.

Let $V(t) = t^2 - 30t + 600$, where $V(t)$ represents the number of gallons of water in a tank t minutes after a drain is opened. Find the average rate of change over $[8, 11]$.

a.) $V(8)$ (2)

424

b.) Which sentence correctly states the result **with units**? (1)

(D) Between $t = 8$ and $t = 11$ minutes, the amount of water in the tank changes at an average rate of -11 gallons per minute.

3 pts

WORKED SOLUTION

$$V(8) = 424, \quad V(11) = 391$$

$$\frac{V(11) - V(8)}{11 - 8} = \frac{391 - (424)}{3} = \frac{-33}{3} = -11 \text{ gallons per minute}$$

It's an *average* over an interval (not an instant), and the units are gallons per minute — gallons per minute. The total change would be -33 gallons.

11.

Let $f(x) = \frac{6}{x}$. Calculate the average rate of change of f over each interval. (Rounded answers to 2+ decimal places are fine.)

a.) $[3, 5]$ (2)

$$-\frac{2}{5}$$

b.) As the interval shrinks toward $x = 3$, these values approach... (1)

$$(C) -\frac{2}{3}$$

3 pts

WORKED SOLUTION

$$[3, 5] : \frac{f(5) - f(3)}{2} = -\frac{2}{5} \approx -0.4$$

$$[3, 4] : \frac{f(4) - f(3)}{1} = -\frac{1}{2} \approx -0.5$$

$$[3, 3.1] : \frac{f(3.1) - f(3)}{0.1} = -\frac{20}{31} \approx -0.6452$$

$$[3, 3.01] : \frac{f(3.01) - f(3)}{0.01} = -\frac{200}{301} \approx -0.6645$$

Each value is the slope of a secant line through $(3, f(3))$. As the interval shrinks they approach the slope of the **tangent line**: $f'(x) = -\frac{6}{x^2}$, so $f'(3) = -\frac{2}{3}$.

12.

Let $f(x) = -x^2 - 3x - 5$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it to fill in the table.

a.) Simplified difference quotient (in terms of x and h) (3)

$$-2x - h - 3$$

b.) As $h \rightarrow 0$, what value do these approach? (2)

$$3$$

5 pts

WORKED SOLUTION

$$f(x+h) = -(x+h)^2 - 3(x+h) - 5 = -x^2 - 2xh - h^2 - 3x - 3h - 5$$

$$f(x+h) - f(x) = -2xh - h^2 - 3h$$

$$\frac{f(x+h) - f(x)}{h} = \frac{h(-2x - h - 3)}{h} = -2x - h - 3$$

Now substitute:

$$x = -3, h = 1 : \quad 2$$

$$x = -3, h = 0.1 : \quad \frac{29}{10} = 2.9$$

$$x = -3, h = 0.01 : \quad \frac{299}{100} = 2.99$$

Letting $h \rightarrow 0$ leaves **3**: the instantaneous rate of change (slope of the tangent line) at $x = -3$. That's exactly what section 1.4 formalizes.

13.

Let $f(x) = \frac{2}{x}$. Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$.

a.) $f'(x) =$ (3)

$$\frac{-2}{x^2}$$

b.) What does $f'(-2) = -\frac{1}{2}$ mean? (1)

(C) The instantaneous rate of change of f at $x = -2$ is $-\frac{1}{2}$ — the slope of the tangent line there.

4 pts

WORKED SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\frac{2}{x+h} - \frac{2}{x}}{h} = \lim_{h \rightarrow 0} \frac{\frac{2x - 2(x+h)}{x(x+h)}}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2h}{x(x+h)} \cdot \frac{1}{h} = \lim_{h \rightarrow 0} \frac{-2}{x(x+h)} = \frac{-2}{x^2} \end{aligned}$$

Then $f'(-2) = -\frac{1}{2}$ and the slope of the tangent at $x = 2$ is $f'(2) = -\frac{1}{2}$.

$f'(-2)$ is the *instantaneous* rate of change at one input — the slope of the tangent line, not the height of the graph and not an average over an interval.

14.

Let $f(x) = 2x^2 + 5x + 5$. Find the equation of the tangent line to the graph of f at the point $(2, 23)$.

a.) $f'(x) =$ (3)
 $4x + 5$

b.) Tangent line (3)
 $y = 13x - 3$

6 pts

WORKED SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{[2(x+h)^2 + 5(x+h) + 5] - (2x^2 + 5x + 5)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 + 5x + 5h + 5 - 2x^2 - 5x - 5}{h} \\ &= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 + 5h}{h} = \lim_{h \rightarrow 0} \frac{h(4x + 2h + 5)}{h} \\ &= \lim_{h \rightarrow 0} (4x + 2h + 5) = 4x + 5 \end{aligned}$$

Slope: $m = f'(2) = 13$.

$$y - 23 = 13(x - 2) \Rightarrow y = 13x - 3$$

Answers in point-slope form are accepted too.

15. Let $f(x) = x^2 - 3$. Complete the limit definition, then interpret $f'(3)$. 4 pts

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{((x+h)^2 - 3) - (x^2 - 3)}{h} = ?$$

a.) $f'(x) =$ (3)

$$2x$$

b.) What does $f'(3) = 6$ tell you? (1)

(A) The slope of the tangent line to the graph of f at $x = 3$ is 6.

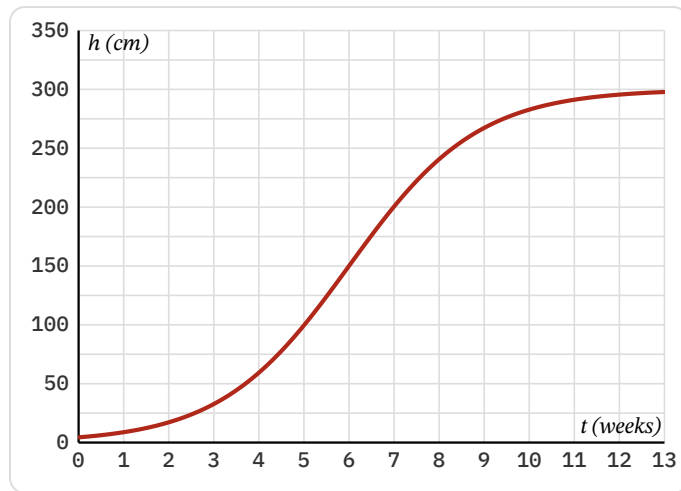
WORKED SOLUTION

$$\lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 3 - x^2 + 3}{h} = \lim_{h \rightarrow 0} \frac{h(2x + h)}{h} = \lim_{h \rightarrow 0} (2x + h) = 2x$$

$f'(3) = 2(3) = 6$. The graph of f has an instantaneous rate of change of 6 at $x = 3$:
the tangent line there has slope 6 (f is increasing).

16.

A sunflower seedling is planted and its height h (in centimeters) after t weeks is graphed below. Decide whether each statement is true or false.



- a.) The slopes of the tangent lines get steeper up to $t = 3$, and then the slopes start to flatten out. (1)

False

- b.) The average rate of change over $[7, 9]$ is less than over $[2, 4]$. (1)

False

2 pts

WORKED SOLUTION

a. **False.** The graph is steepest at about $t = 6$ (where it changes from bending up to bending down).

b. **False.** $\text{ARC} \approx \frac{295.6 - 4.4}{12} \approx 24.3$ cm/week; the tangent slope at $t = 7$ is about 46.6 cm/week.

c. **True.** Tangent slopes: about 46.6 cm/week at $t = 5$ vs. 6 at $t = 1$.

d. **False.** $[7, 9]: \approx 33.4$ cm/week; $[2, 4]: \approx 21.1$ cm/week.
